

ZEMETCHINO, Russian Federation

WMO: 278570

Lat: 53.483N

Lon: 42.633E

Elev: 131

StdP: 99.76

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.3	-24.1	-29.8	0.2	-27.2	-26.5	0.3	-23.9	8.9	-4.8	8.1	-3.8	1.1	150	0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	32.2	21.0	30.0	19.7	28.1	19.0	22.1	29.9	20.9	27.9	19.8	26.2	2.7	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.4	14.4	25.9	18.4	13.4	24.3	17.4	12.7	23.1	65.3	30.1	60.9	27.9	57.3	26.2	26.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.5	6.6	5.6	DB	-30.8	34.3	3.3	2.4	-33.2	36.0	-35.1	37.4	-36.9	38.7	-39.3	40.4	(4)
(5)			WB	-30.9	23.3	3.2	1.3	-33.3	24.2	-35.2	25.0	-37.0	25.7	-39.3	26.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.8	-8.9	-8.3	-3.1	6.9	14.3	18.1	20.4	18.2	12.5	6.0	-0.9	-6.3	(6)	
	DBStd	11.77	7.49	7.21	5.25	5.11	4.85	3.87	3.56	4.08	4.19	4.42	5.44	6.85	(7)	
	HDD10.0	2632	587	512	407	122	15	1	0	0	21	137	327	504	(8)	
	HDD18.3	4764	846	745	665	344	143	51	19	52	178	381	577	763	(9)	
	CDD10.0	1108	0	0	0	27	148	243	322	256	97	14	0	0	(10)	
	CDD18.3	199	0	0	0	0	18	44	83	49	5	0	0	0	(11)	
	CDH23.3	2194	0	0	0	8	233	478	853	551	71	0	0	0	(12)	
	CDH26.7	756	0	0	0	1	62	153	309	219	12	0	0	0	(13)	
(14) Wind	WSAvg	2.5	2.8	2.9	2.7	2.6	2.5	2.1	1.9	1.9	2.0	2.5	2.7	2.8	(14)	
(15) Precipitation	PrecAvg	496	34	27	29	28	39	53	73	47	45	43	44	39	(15)	
	PrecMax	755	69	70	74	65	114	171	362	114	106	86	96	86	(16)	
	PrecMin	291	5	1	0	7	3	8	6	1	2	0	5	9	(17)	
	PrecStd	105	17	17	17	17	31	37	65	29	25	22	22	18	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.1	3.0	10.8	24.5	30.9	33.6	36.2	36.6	28.3	20.2	11.5	6.5	(19)	
		MCWB	2.1	1.9	6.0	14.1	18.5	21.7	23.8	24.3	17.2	13.3	9.8	5.6	(20)	
	2%	DB	1.7	2.1	7.4	21.5	28.5	30.2	32.2	31.9	25.3	17.0	9.2	3.9	(21)	
		MCWB	1.0	1.3	4.1	13.1	17.5	19.7	21.4	21.0	16.4	12.2	8.2	3.1	(22)	
	5%	DB	1.1	1.5	4.7	18.7	26.1	28.3	29.9	28.7	22.7	14.5	7.5	2.2	(23)	
		MCWB	0.5	0.8	2.4	11.6	16.6	19.0	20.3	19.4	15.3	11.1	6.5	1.5	(24)	
	10%	DB	0.4	0.8	3.0	15.9	23.5	26.1	27.9	26.0	20.0	12.6	5.7	1.2	(25)	
		MCWB	-0.1	0.3	1.4	10.1	15.4	18.3	19.7	18.1	14.4	10.2	4.7	0.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.5	2.2	6.8	15.6	19.8	23.0	23.9	24.5	18.7	14.9	10.3	5.7	(27)	
		MCDB	2.7	2.6	10.7	21.6	27.5	31.6	34.1	36.1	24.6	18.1	11.1	6.2	(28)	
	2%	WB	1.2	1.5	4.3	13.6	18.4	20.9	22.4	21.6	17.5	13.0	8.4	3.1	(29)	
		MCDB	1.5	1.9	6.8	20.0	26.5	27.9	30.3	29.6	22.9	15.4	9.1	3.8	(30)	
	5%	WB	0.6	0.9	2.7	12.1	17.2	19.8	21.3	20.1	16.2	11.6	6.6	1.7	(31)	
		MCDB	1.0	1.3	4.3	17.6	24.4	26.3	28.2	27.0	21.2	14.1	7.4	2.1	(32)	
	10%	WB	-0.1	0.3	1.7	10.6	16.0	18.9	20.2	18.9	14.9	10.2	4.8	0.8	(33)	
		MCDB	0.5	0.8	2.8	15.4	22.7	24.8	26.4	24.6	19.3	12.3	5.6	1.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.3	7.2	7.9	10.3	11.7	10.7	11.1	11.4	10.0	7.4	5.0	5.3	(35)	
		MCDBR	3.9	4.9	8.8	14.6	15.1	14.7	14.6	15.9	15.0	11.3	6.6	4.1	(36)	
	5% WB	MCWBR	3.9	4.6	6.0	7.8	6.8	6.3	6.1	6.7	7.2	7.0	5.6	3.8	(37)	
		MCWBR	3.9	4.7	7.2	13.6	13.6	12.7	12.7	14.1	12.3	10.1	5.8	3.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.295	0.331	0.358	0.426	0.392	0.387	0.412	0.432	0.379	0.358	0.329	0.298	(40)	
		taud	2.097	2.027	2.043	2.091	2.313	2.353	2.295	2.224	2.352	2.327	2.228	2.105	(41)	
	Ebn at Noon		639	732	804	797	856	863	832	786	788	714	603	551	(42)	
		Edh at Noon	71	106	129	142	120	116	121	123	96	80	64	59	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.83	1.75	3.33	4.24	5.61	5.78	5.83	4.79	3.06	1.51	0.63	0.46	(44)	
	RadStd		0.16	0.30	0.44	0.42	0.49	0.56	0.53	0.43	0.40	0.22	0.15	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.63	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+116	N/A
(47) Regional (5 neighbors)		+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+131	N/A

Nomenclature: See separate page